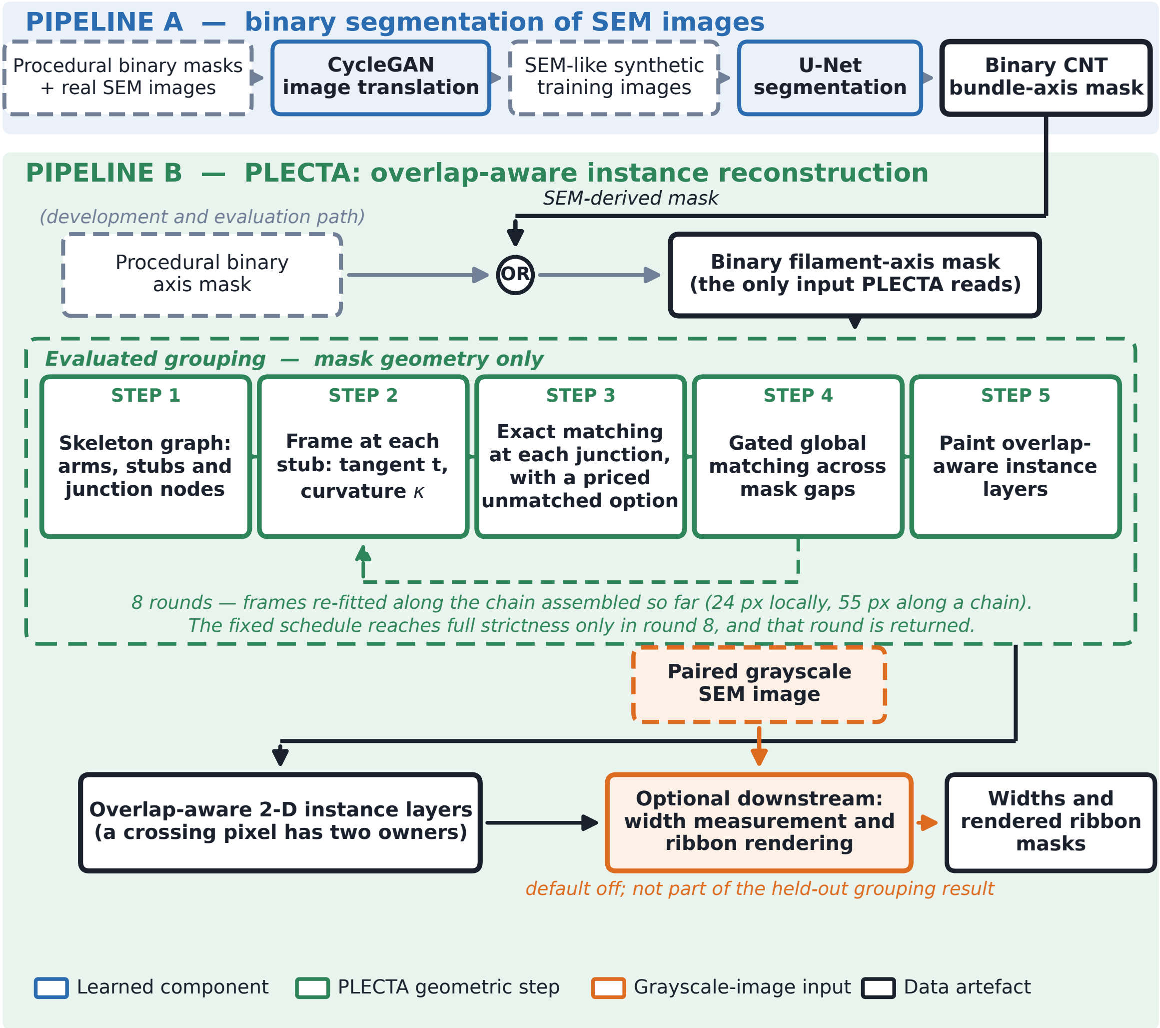
PIPELINE A — binary segmentation of SEM images
Procedural binary masks + real SEM images
CycleGAN image translation
SEM-like synthetic training images
U-Net segmentation
Binary CNT bundle-axis mask
PIPELINE B — PLECTA: overlap-aware instance reconstruction
SEM-derived mask
(development and evaluation path)
Procedural binary axis mask
OR
Binary filament-axis mask (the only input PLECTA reads)
Evaluated grouping — mask geometry only
STEP 1
Skeleton graph: arms, stubs and junction nodes
STEP 2
Frame at each stub: tangent t, curvature κ
STEP 3
Exact matching at each junction, with a priced unmatched option
STEP 4
Gated global matching across mask gaps
STEP 5
Paint overlap-aware instance layers
8 rounds — frames re-fitted along the chain assembled so far (24 px locally, 55 px along a chain).
The fixed schedule reaches full strictness only in round 8, and that round is returned.
Paired grayscale SEM image
Overlap-aware 2-D instance layers (a crossing pixel has two owners)
Optional downstream: width measurement and ribbon rendering
Widths and rendered ribbon masks
default off; not part of the held-out grouping result
Learned component
PLECTA geometric step
Grayscale-image input
Data artefact